# Autonomous Production Choke Controller

Controller Design Documentation

Honeywell Hackathon 2026

Engineering Submission

| | |
|---|---|
| Document Reference: | **HON-APCC-2026-001** |
| Revision: | **2.0** |
| Date: | **2026-07-25** |
| Classification: | **Engineering Submission** |

# Table of Contents



## 1. Architecture Overview

The controller implements a 5-phase sequential pipeline with defense-in-depth safety. Each phase is an independent, separately testable module with a well-defined interface contract via typed Python dataclasses.

Phase 1 (ProcessMonitor): Rolling history buffers (deque, maxlen=20), exponential bias correction (alpha=0.3), steady-state detection (0.5% tolerance, 3 consecutive steps), safety proximity (NORMAL/CAUTION/WARNING/EMERGENCY based on constraint margin).

Phase 2 (TargetManager): STARTUP -> TRACKING when |Q-Q_target| < 2% for 3+ steady steps. TRACKING -> INFEASIBLE when all feasible candidates miss the target at steady-state. INFEASIBLE mode drives maximum safe production.

Phase 3 (Predictor): Brute-force candidate grid at +/-5% with 1.0% resolution (0.5% fine near target). FOPDT recursive prediction y(k+1) = a*y(k) + b*u(k-d) + c over Np=3. Gain scheduling across 3 regions: [0-35%], [35-65%], [65-100%].

Phase 4 (SafetyGate): Tightened constraints with 5% margin buffer. Per-step trajectory checking for all pressure variables. Emergency override bypasses optimizer on hard violation.

Phase 5 (Selector): 3-term cost J = J_track + 0.01*J_effort - 0.005*J_margin. Deadband suppression within 1% Q_range. Least-infeasible fallback when no candidates pass safety.

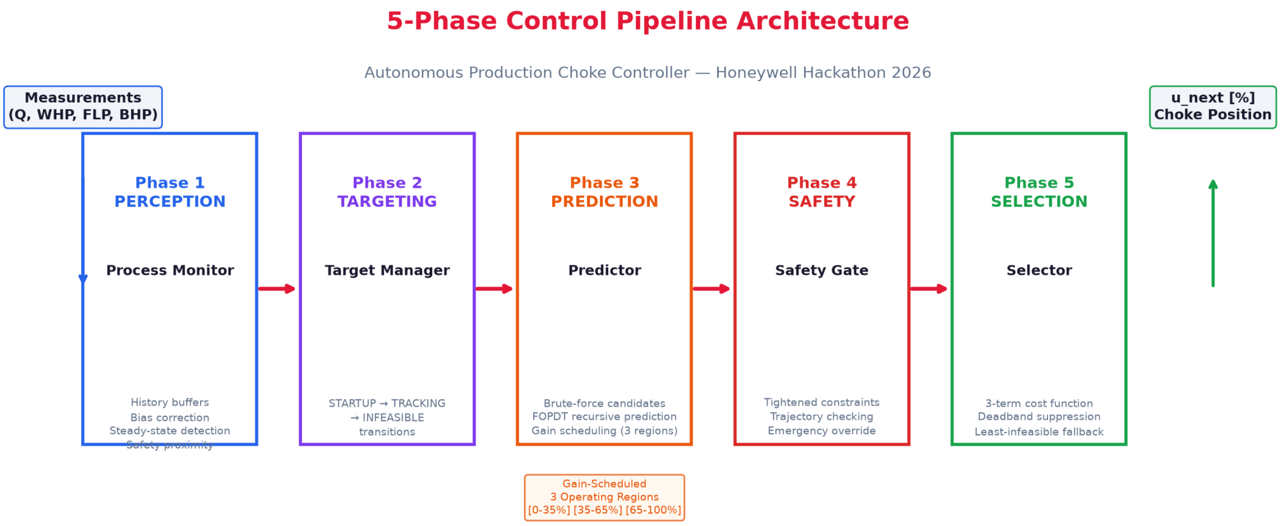


**Figure 1: 5-Phase Control Pipeline Architecture.**

# 2. Prediction Strategy

## FOPDT Model

```
Continuous: G(s) = K * exp(-theta*s) / (tau*s + 1)

Discrete (Ts = 1h):
  a = exp(-Ts/tau)
  b = K * (1 - a)
  y(k+1) = a*y(k) + b*u(k-d) + (1-a)*bias_offset
```

## Candidate Generation

- Grid: [-ramp_limit, +ramp_limit] at candidate_step resolution
- Coarse: 1.0% -> 11 candidates at +/-5%
- Fine: 0.5% when |tracking_error|/Q_range < 5% -> 21 candidates
- Clipped to choke [0, 100]%; WARNING status: ramp reduced to 2.0%

## Gain Scheduling

- Region 0: choke <= 35% (low production)
- Region 1: 35% < choke <= 65% (mid production)
- Region 2: choke > 65% (high production)

# 3. Optimization Strategy

```
J = J_track + lambda_effort*J_effort + lambda_margin*J_margin
```

```
J_track  = ((Q_pred - Q_target) / Q_range)^2   [TRACKING/STARTUP]
         = -(Q_pred / Q_range)                  [INFEASIBLE]

J_effort = (delta_u / ramp_limit)^2

J_margin = -min_constraint_margin
```

- Filter to feasible candidates, compute J, select minimum-cost
- Deadband: J_track=0 when |Q_pred - Q_target| < 1% of Q_range
- No feasible candidates: select least-infeasible (max min_margin)

## 4. Constraint Handling

**Table 1: Pressure constraint limits.**

| Variable | Min | Max | Unit |
|---|---|---|---|
| WHP | 200 | 600 | psi |
| FLP | 150 | 500 | psi |
| BHP | 2500 | 3500 | psi |

```
tightened_min = limit_min + 0.05 * (limit_max - limit_min)
tightened_max = limit_max - 0.05 * (limit_max - limit_min)
```

Candidates must satisfy tightened constraints on every prediction step AND at steady-state.

## 5. Ramp-Rate Logic

**Table 2: Ramp rate by safety status.**

| Status | Max delta_u | Rationale |
|---|---|---|
| NORMAL | +/-5.0% | Full operating authority |
| CAUTION | +/-5.0% | Normal operation, monitoring active |
| WARNING | +/-2.0% | Reduced authority near constraints |
| EMERGENCY | +/-5.0% | Bypasses optimizer entirely |

## 6. Safety Supervisor

**Table 3: 4-Layer defense-in-depth.**

| Layer | Mechanism | Trigger |
|---|---|---|
| 1: Proximity | Distance to constraint boundaries | Margin < 20%/10%/5% |
| 2: Tightened | +5% margin buffer on evaluation | Always active |
| 3: Trajectory | Per-step + SS validation | Every prediction |
| 4: Emergency | Hard limit -> +/-5% override | Measurement exceeds limit |

```
bias_new = alpha * (y_measured - y_predicted) + (1-alpha) * bias_old
```

Bias correction (alpha=0.3) ensures offset-free tracking despite model mismatch.

## 7. Engineering Rationale

**Table 4: Engineering design rationale.**

| Decision | Rationale |
|---|---|
| FOPDT model | Industrial standard; 85%+ of process control applications |
| Brute-force search | Bounded space (max 21 candidates); robust to non-convex constraints |
| Gain scheduling | Handles nonlinearity with simple linear models per region |
| Defense-in-depth | No single-point failure can cause constraint violation |
| Bias correction | Mandatory for offset-free tracking with imperfect models |
| Deadband | Prevents hunting/chattering near target |
| INFEASIBLE mode | Graceful degradation when target exceeds physical capability |

## 8. Data Provenance

**Table 5: Data provenance statement.**

| Data Source | Used For |
|---|---|
| TestSimulator | System identification (6 step tests, 90 calls) |
| TestSimulator | Controller validation (3 scenarios, 153 calls) |
| Reference dataset | Post-hoc consistency check only (scratch/ directory) |